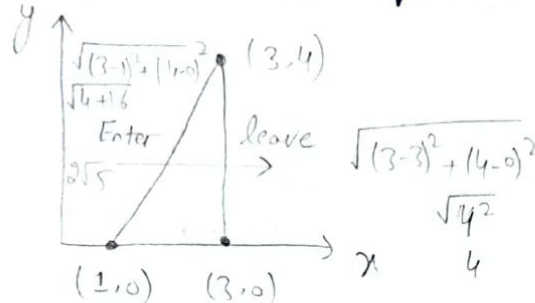


Applications of Multiple Integrals

Q:- The population density of Gladstone City is given by $P(x,y) = 2x^2 + 5y$, where x & y are in miles & P is the no. of people per square mile, in thousands. The city limits are as shown in the graph below.

(a) Determine the city's population

(b) Determine the average no. of people per square mile of the city.



Sol
$$\int_c^b \int_a^d P(x,y) dx dy$$

$$\frac{\sqrt{(3-1)^2 + (0-0)^2}}{\sqrt{(2)^2}} = \frac{2}{2} = 1$$

Since we will integrate wrt ' x ' first, we ~~will~~ imagine an arrow going in a +ve x -direction. The 1st line that our arrow touches will represent our lower bound of the inner integral. That is,

$$g(y) = my + b$$

For this we will calculate the slope & x -intercept

$$m = \frac{x_2 - x_1}{y_2 - y_1} = \frac{3-1}{4-0} = \frac{2}{4} = \frac{1}{2}$$

$$g(y) = \frac{1}{2}y + b$$

x -intercept

$$1 = 0 + b$$

$$\boxed{b = 1}$$

Lower bound is $g(y) = \frac{y}{2} + 1$ & upper bound is vertical line $x = 3$.

For outer integral the bounds are constants, the region expands from $y = 0$ to $y = 4$

$$\int_0^4 \int_{\frac{y}{2}+1}^3 [2x^2 + 5y] dx dy$$

$$\int_0^4 \left[\frac{2x^3}{3} + 5yx \right]_{\frac{y}{2}+1}^3 dy$$

$$\int_0^4 - \frac{(y+4)(y^2+4y+52)}{12} dy$$

$$- \frac{y^4}{48} - y^3 + \frac{9y^2}{2} + \frac{52y}{3} \Big|_0^4$$

$$72$$

The city's population is 72 people per square mile

Since the region of integration R is a triangle with vertices $(1,0)$, $(3,0)$ & $(3,4)$. Its area can be found using the formula for the area of triangle $A = \frac{1}{2}bh$

$$A(R) = \frac{1}{2} (2)(4)$$

$$A(R) = 4$$

$$P_{av} = \frac{1}{A(R)} \int_0^4 \int_{\frac{y}{2}+1}^3 [2x^2+5y] dx dy$$

$$\approx \frac{1}{4} (72)$$

$$\approx 18$$

The average no. of people per sq mile of the city is 18.

Q:- The total force on the dam is given by surface integral

$$\int_A K(h-y) dA$$

Imagine that the dam is rectangular in profile with a width of 100m & a height h of 40m. The constant ' K ' is assumed to be $10^4 \text{ kgm}^{-2} \text{ s}^{-2}$. Construct the double integral & then use it to find total force on dam.

Sol $40 \int_0^{100} \int_0^{100} 10^4 (40 - y) dx dy$

$$40 \int_0^{100} 10^4 (40 - y) x \Big|_0^{100} dy$$

$$100 \times 10^4 \int_0^{40} (40 - y) dy$$

$$10^6 \left[40y - \frac{y^2}{2} \right]_0^{40}$$

$$10^6 \left[(40 \times 40 - \frac{40^2}{2}) - 0 \right]$$

$$10^6 \times 800$$

$$8 \times 10^8 \text{ N}$$

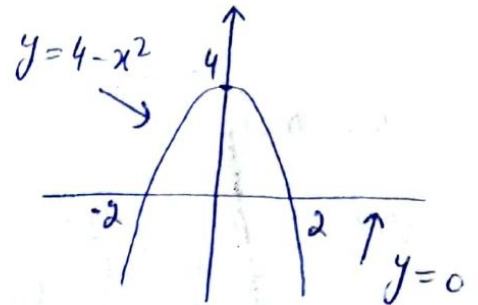
The total force is 800 meganewtons.

Total Electric Charge

Q:- A storage bin is to be constructed in the shape of the solid bounded above by the surface $z = 20 - x^2 - y^2$ below by the xy plane, & on the sides by the plane $y = 0$ & the parabolic cylinder $y = 4 - x^2$, where x & y are in meters. Find the vol of bin.

Sol

$$V = \iint_R f(x, y) dA$$



We have to find the volume (V) of the fn $f(x, y)$ over the region R bounded by the plane $y = 0$ & parabolic cylinder $y = 4 - x^2$

$$4 - x^2 = 0$$

$$4 = x^2$$

$$x = \pm 2$$

$$R : -2 \leq x \leq 2, 0 \leq y \leq 4 - x^2$$

$$V = \int_{-2}^2 \int_0^{4-x^2} (20 - x^2 - y^2) dy dx$$

$$= \int_{-2}^2 \left[20y - x^2y - \frac{y^3}{3} \right]_0^{4-x^2} dx$$

$$= \int_{-2}^2 \left(\frac{1}{3}x^6 - 3x^4 - 8x^2 + \frac{176}{3} \right) dx$$

$$= \left[\frac{1 \cdot x^7}{3 \cdot 7} - \frac{3x^5}{5} - \frac{8x^3}{3} + \frac{176}{3}x \right]_{-2}^2$$

$$V = 165.79 \text{ m}^3$$

Q:- A fancy jewellery box has the shape of the solid bounded above by the plane $3x+4y+2z=12$ below by the xy plane, & on the sides by the planes $x=0$ & $y=0$, where x, y & z are in inches. Find the vol of box.

Sol

$$2z = 12 - 3x - 4y$$

$$z = 6 - \frac{3}{2}x - 2y$$

The trace in xy -plane is the line $\frac{3}{2}x + 2y = 6$ or $y = -\frac{3}{4}x + 3$

$$\text{Vol} = 4 \int_0^4 \int_0^{-\frac{3}{4}x+3} (6 - \frac{3}{2}x - 2y) dy dx$$

$$= 4 \int_0^4 \left(6y - \frac{3}{2}xy - y^2 \right) \Big|_0^{-\frac{3}{4}x+3} dx$$

$$= 4 \int_0^4 \left[6 \left(-\frac{3}{4}x + 3 \right) - \frac{3}{2}x \left(-\frac{3}{4}x + 3 \right) - \left(-\frac{3}{4}x + 3 \right)^2 \right] dx$$

$$= 4 \int_0^4 \frac{9}{16} (x^2 - 8x + 16) dx$$

$$= \frac{9}{16} \left(\frac{1}{3}x^3 - 4x^2 + 16x \right) \Big|_0^4$$

$$= \frac{9}{16} \left[\frac{1}{3}(4^3) - 4(4)^2 + 16(4) \right]$$

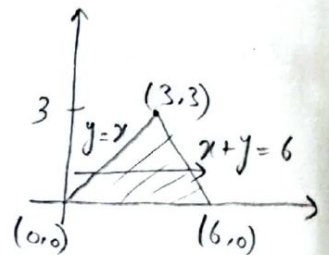
- 19

Q:- A biomass covers the triangular bottom of the container whose region is bounded by the x -axis, & the lines $y = x$ & $x + y = 6$.
The depth $h(x, y) = \frac{x}{y+2}$ at each pt (x, y) in the region is given.

All the dimensions are in cm. What is the total volume of biomass?
sol Vol is given by double integral $V = \iint_R h(x,y) dA$

Region is bounded by x -axis ($y=0$) & lines $x=y$ & $x=6-y$

$$V = \int_0^3 \int_y^{6-y} \frac{x}{y+2} dx dy$$



$$= \int_0^3 \frac{1}{y+2} \cdot \frac{x^2}{2} \Big|_y^{6-y} dy$$

$$= \int_0^3 \frac{1}{2(y+2)} [(6-y)^2 - y^2] dy$$

$$= \int_0^3 \frac{1}{2(y+2)} [36 - 12y] dy$$

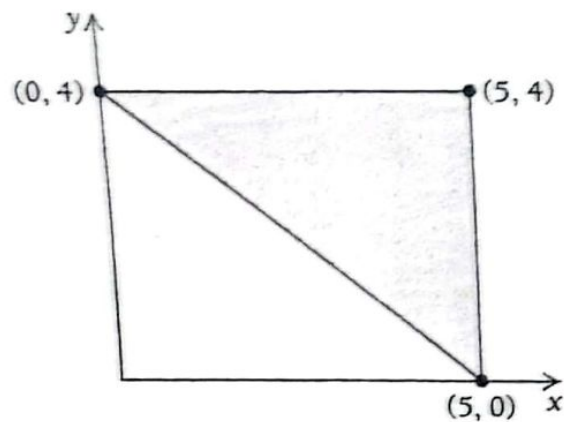
$$= \int_0^3 \left[-6 + \left(\frac{60}{2y+4} \right) \right] dy$$

$$= -6y + 30 \ln|2y+4| \Big|_0^3$$

$$= [-6(3) + 30 \ln(2(3)+4)] - [-6(0) + 30 \ln(0+4)]$$

$$\approx 9.489$$

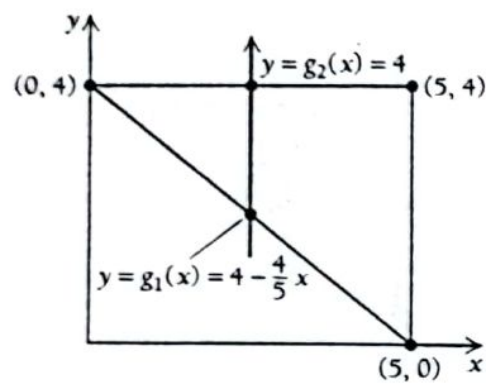
Vol of biomass is approximately 9.5 cm^3



Business: Demographics and Vehicle Ownership. The density of privately owned vehicles in Leatonville is given by $p(x, y) = \frac{1}{8}xy$, where x is miles in the east–west direction, y is miles in the north–south direction, and p is the number of privately owned vehicles per square mile, in thousands. If the city limits are as shown in the figure to the left, what is the total number of privately owned vehicles in Leatonville?

Solution If we choose to integrate with respect to y first and x second, the iterated integral is

$$\int_a^b \int_{g_1(x)}^{g_2(x)} \frac{1}{8}xy \, dy \, dx. \quad \text{See below for clarification of } g_1(x) \text{ and } g_2(x).$$



Since we are integrating with respect to y first, a helpful visual method for determining the bounds of integration is to draw an arrow in the positive y -direction, intersecting the region representing the city. The arrow enters the region at $g_1(x) = 4 - \frac{4}{5}x$ and exits the region through $g_2(x) = 4$. The bounds for the outer integral, with respect to x , are constants: the region extends from $x = 0$ to $x = 5$.

The diagonal boundary is a line with slope $-\frac{4}{5}$ and a y -intercept of 4. Therefore, $g_1(x) = 4 - \frac{4}{5}x$. The horizontal boundary is $g_2(x) = 4$. The integral is thus

$$\int_0^5 \int_{4-\frac{4}{5}x}^4 \frac{1}{8}xy \, dy \, dx.$$

We integrate the inside integral first, with respect to y :

$$\begin{aligned} \int_{4-\frac{4}{5}x}^4 \frac{1}{8}xy \, dy &= \frac{1}{8}x \left[\frac{1}{2}y^2 \right]_{4-\frac{4}{5}x}^4 \\ &= \frac{1}{8}x \left[\frac{1}{2}(4)^2 - \frac{1}{2}\left(4 - \frac{4}{5}x\right)^2 \right] \\ &= \frac{1}{8}x \left(\frac{16}{5}x - \frac{8}{25}x^2 \right) \\ &= \frac{2}{5}x^2 - \frac{1}{25}x^3. \end{aligned} \quad \left. \vphantom{\int_{4-\frac{4}{5}x}^4} \right\} \text{Simplifying}$$

We now integrate with respect to x :

$$\begin{aligned} \int_0^5 \left(\frac{2}{5}x^2 - \frac{1}{25}x^3 \right) dx &= \left[\frac{2}{15}x^3 - \frac{1}{100}x^4 \right]_0^5 \\ &= \left(\frac{2}{15}(5)^3 - \frac{1}{100}(5)^4 \right) - 0 \\ &= \frac{125}{12} \approx 10.417. \end{aligned}$$

Quick Check 2

Repeat Example 2, integrating with respect to x first and y second.

Therefore, Leatonville has about 10,417 privately owned vehicles.

Business: Average Vehicle Ownership. Find the average number of privately owned vehicles per square mile in Leatonville (see Example 2).

Solution Since the region of integration R is a triangle with vertices $(5, 0)$, $(5, 4)$, and $(0, 4)$, its area can be found using the formula for the area of a triangle, $A = \frac{1}{2}bh$. Thus, we have

$$A(R) = \frac{1}{2}(5)(4) = 10.$$

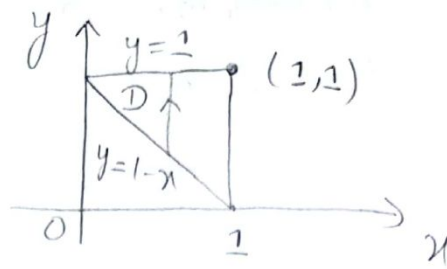
Therefore, the average number of privately owned vehicles per square mile in the city is

$$\begin{aligned} p_{av} &= \frac{1}{A(R)} \int_0^5 \int_{4-\frac{4}{5}x}^4 \frac{1}{8} xy \, dy \, dx \\ &\approx \frac{1}{10} \cdot 10.417 \quad \text{Using the result from Example 2} \\ &\approx 1.042. \end{aligned}$$

1. Leatonville has an average of about 1042 privately owned vehicles per square mile.

3 ✓

Q:- Charge is distributed over the ~~rect~~^{tri}angular region D so that the charge density at (x, y) is $\sigma(x, y) = xy$, measured in Coulombs per square meter (C/m^2). Find total charge.



of mass of
triangular
Mass

Sol

$$Q = \iint_D \sigma(x, y) dA$$

$$Q = \int_0^1 \int_{1-x}^1 xy dy dx$$

$$= \int_0^1 x \left[\frac{y^2}{2} \right]_{1-x}^1 dx$$

$$= \int_0^1 \frac{x}{2} [1 - (1-x)^2] dx$$

$$= \frac{1}{2} \int_0^1 (2x^2 - x^3) dx$$

$$= \frac{1}{2} \left[\frac{2x^3}{3} - \frac{x^4}{4} \right]_0^1$$

$$= \frac{5}{24}$$

Thus total charge is $\frac{5}{24} C$.

7. In a certain factory, output is given by the Cobb-Douglas production function

$$Q(K, L) = 50K^{3/5}L^{2/5}$$

where K is the capital investment in units of \$1,000 and L is the size of the labor force measured in worker-hours. Suppose that monthly capital investment varies between \$10,000 and \$12,000, while monthly use of labor varies between 2,800 and 3,200 worker-hours. Find the average monthly output for the factory.

Solution

It is reasonable to estimate the average monthly output by the average value of $Q(K, L)$ over the rectangular region $R: 10 \leq K \leq 12, 2,800 \leq L \leq 3,200$. The region has area

$$A = \text{area of } R = (12 - 10) \times (3,200 - 2,800) \\ = 800$$

so the average output is

$$\begin{aligned} \text{AV} &= \frac{1}{800} \iint_R 50K^{3/5}L^{2/5} dA \\ &= \frac{1}{800} \int_{2,800}^{3,200} \left(\int_{10}^{12} 50K^{3/5}L^{2/5} dK \right) dL \\ &= \frac{1}{800} \int_{2,800}^{3,200} 50L^{2/5} \left(\frac{5}{8} K^{8/5} \right) \Big|_{K=10}^{K=12} dL \\ &= \frac{1}{800} (50) \left(\frac{5}{8} \right) \int_{2,800}^{3,200} L^{2/5} (12^{8/5} - 10^{8/5}) dL \\ &= \frac{1}{800} (50) \left(\frac{5}{8} \right) (12^{8/5} - 10^{8/5}) \left(\frac{5}{7} L^{7/5} \right) \Big|_{L=2,800}^{L=3,200} \\ &= \frac{1}{800} (50) \left(\frac{5}{8} \right) \left(\frac{5}{7} \right) (12^{8/5} - 10^{8/5}) [(3,200)^{7/5} - (2,800)^{7/5}] \\ &\approx 5,181.23 \end{aligned}$$

Thus, the average monthly output is approximately 5,181 units.

Q:- The total force on the dam is given by surface integral

$$\int_A K(h-y) dA$$

Imagine that the dam is rectangular in profile with a width of 100m & a height h of 40m. The constant 'K' is assumed to be $10^4 \text{ kgm}^{-2}\text{s}^{-2}$. Construct the double integral & then use it to find total force on dam.

Sol

$$40 \int_0^{100} \int_0^{40} 10^4 (40-y) dx dy$$

$$40 \int_0^{100} 10^4 (40-y) x \Big|_0^{100} dy$$

$$100 \times 10^4 \int_0^{40} (40-y) dy$$

$$10^6 \left[40y - \frac{y^2}{2} \right]_0^{40}$$

$$10^6 \left[(40 \times 40 - \frac{40^2}{2}) - 0 \right]$$

$$10^6 \times 800$$

$$8 \times 10^8 \text{ N}$$

The total force is 800 meganewtons.